

Data visualizations

INTRODUCTION TO DATA LITERACY

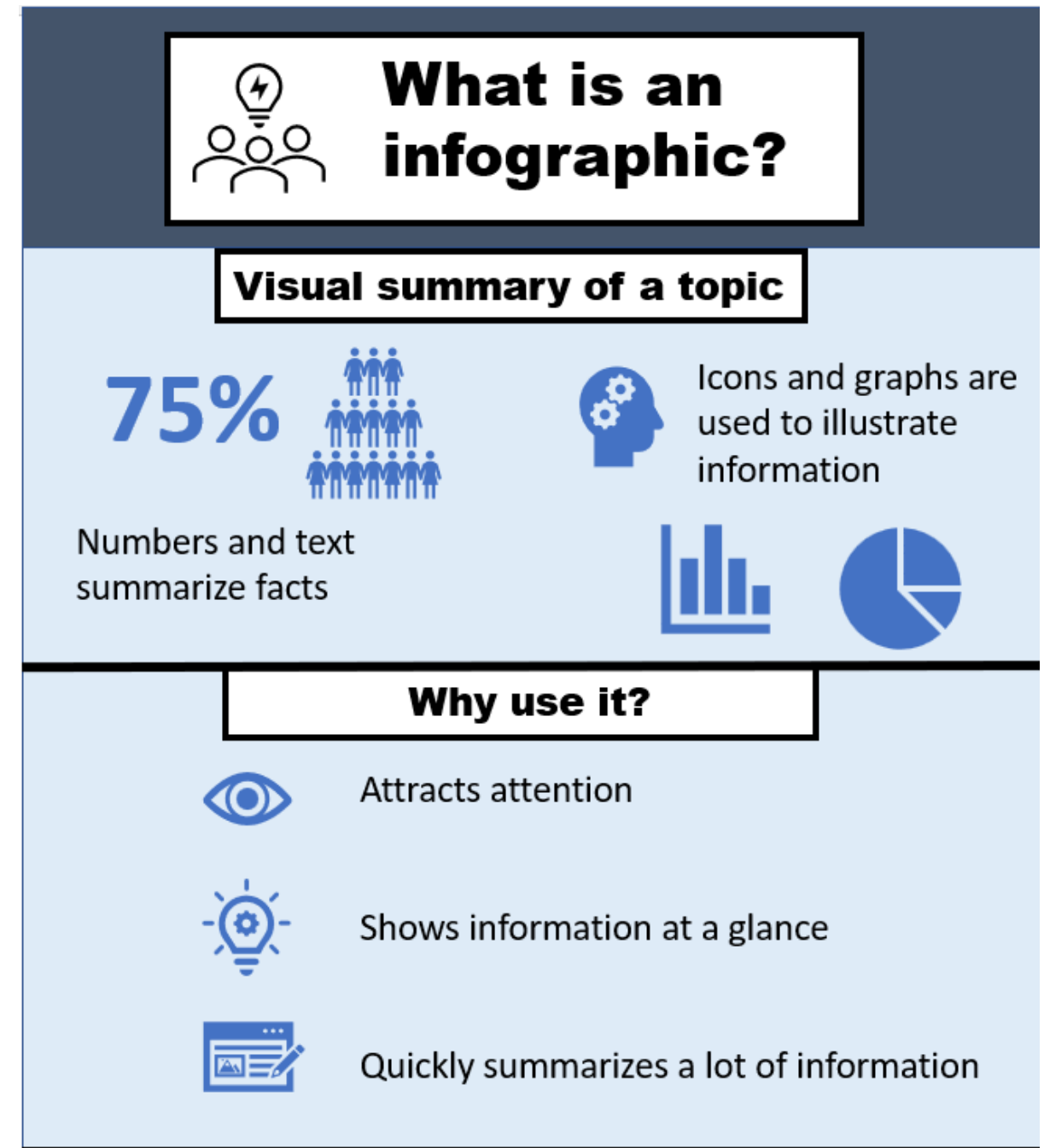


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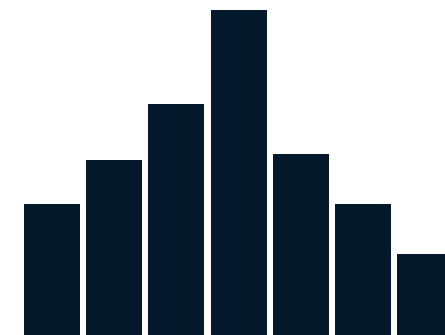
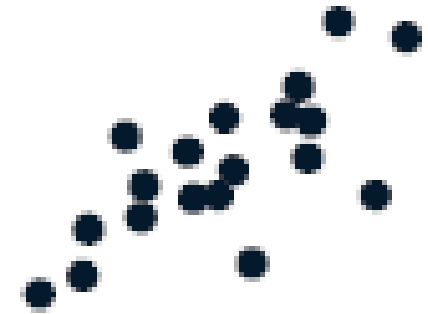
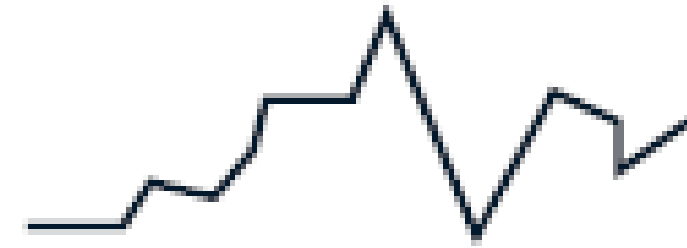
A picture tells a thousand words

- Visualization helps to understand complex data at a glance
- Makes it easier to understand insights identified in the data
- Several data tools and methods rely on visualization:
 - Exploratory data analysis
 - Dashboard
 - Infographic



Common types of visualizations

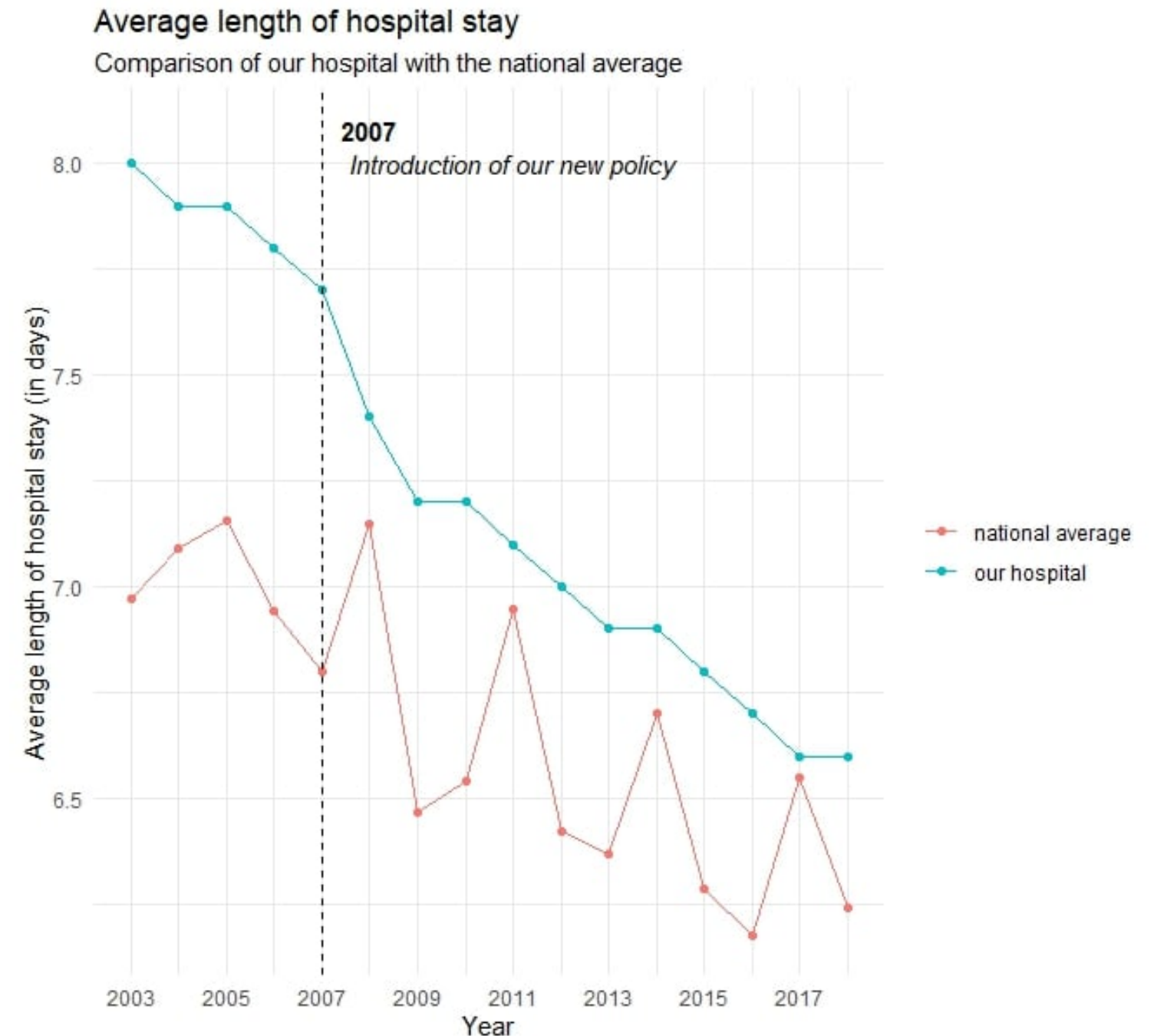
- Capture a trend
 - Line chart
- Visualize relationships
 - Bar chart, scatter plot
- Visualize distributions
 - Histogram, box plot



¹ <https://www.datacamp.com/cheat-sheet/data-viz-cheat-sheet>

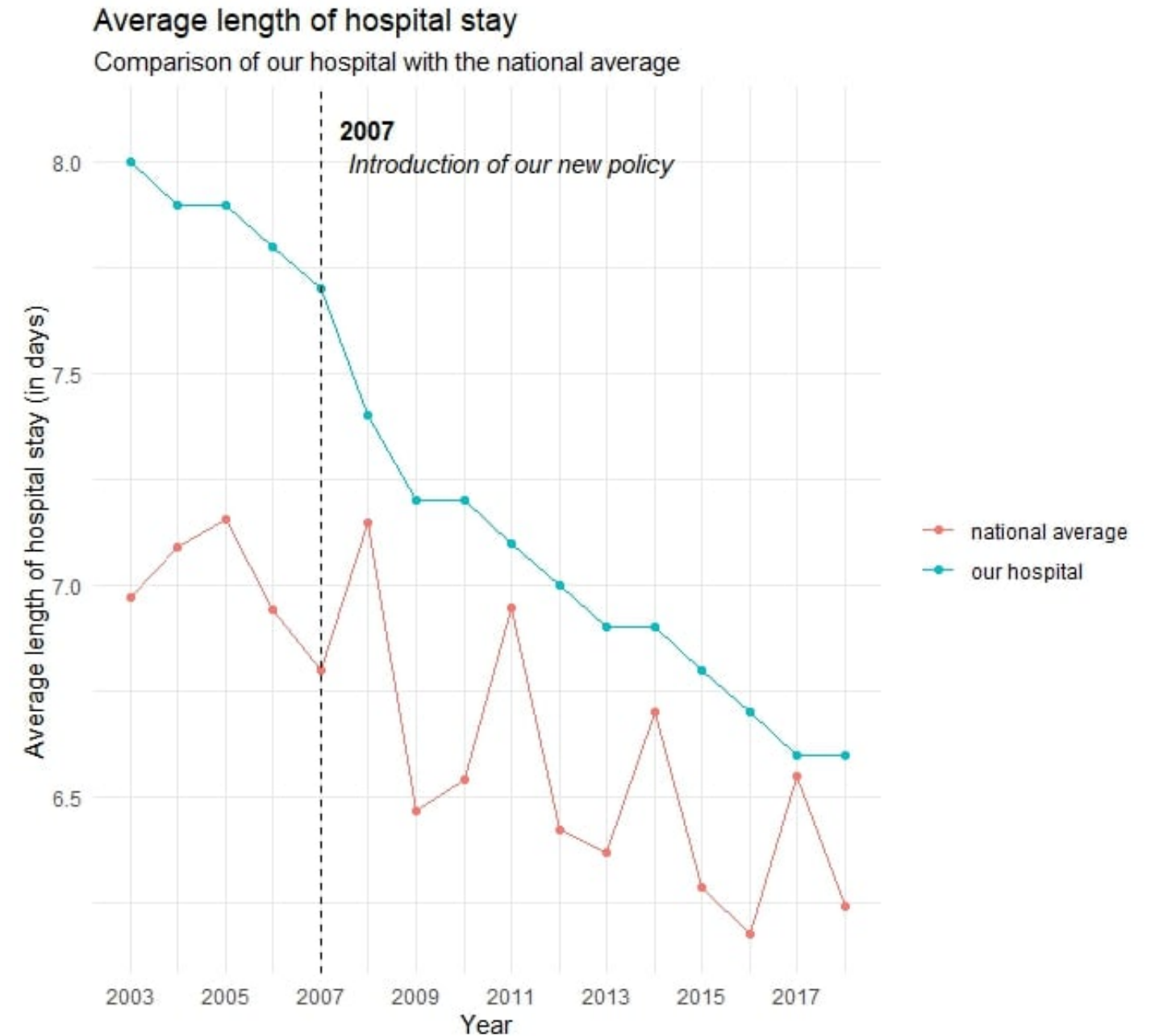
What makes a good visualization?

- Eliminate clutter:
 - Less is more
 - Clarity and readability
- Focus attention:
 - Link to insights
 - Highlight key aspects
- Provide context: use the McCandless technique



McCandless technique

1. Introduce the visualization
2. Anticipate obvious questions
3. State the central insight
4. Provide supporting evidence
5. Closing statement



¹ <https://artscience.blog/home/the-mccandless-method-of-data-presentation>

Let's practice!

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Data storytelling

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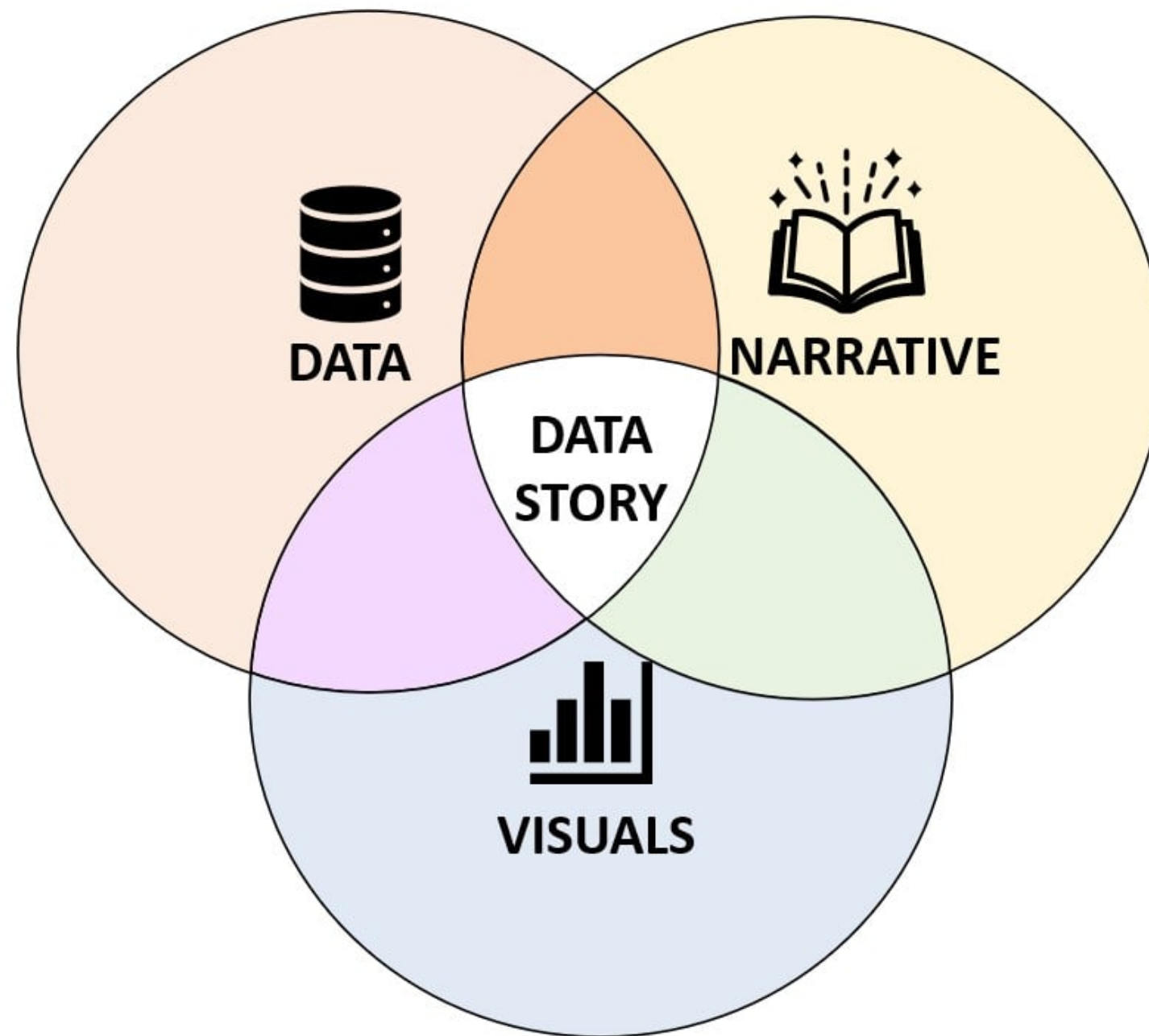
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Why do you need to tell a story?

- Make key points more memorable
- Keep attention and engagement
- Convince your audience of your message
- Motivate them to take action



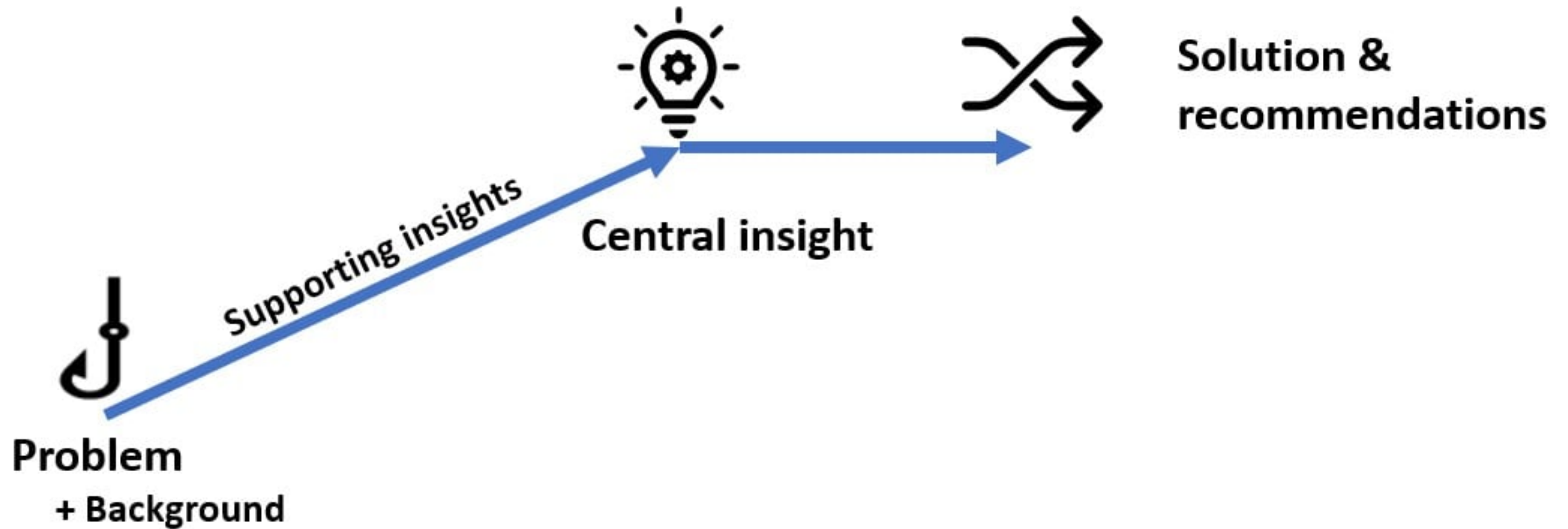
Components of data storytelling



Elements of a narrative

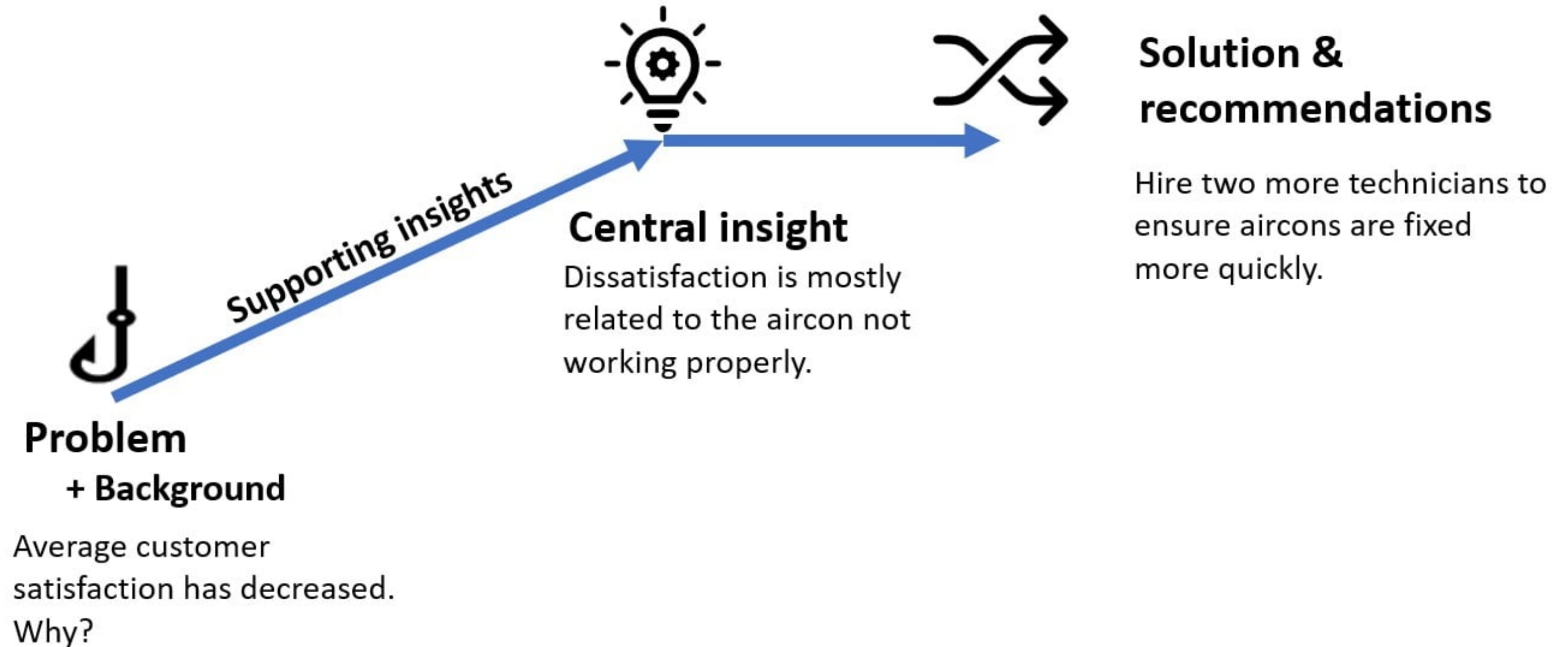
- Characters
 - Problem
 - Setting
 - Plot
 - Resolution
- Stakeholders
 - Business problem
 - Relevant background
 - Central and supporting insights
 - Solution and recommendations

Narrative structure



¹ Dykes, B. (2019). Effective data storytelling. Wiley.

Example narrative



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Three keys to communicating effectively

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The three keys



Focus

Keeps it relevant, to the point and meaningful



Structure

Acts as a guide and increases understandability



Form

Ensures it is tailored to its purpose and the audience

Focus

- Select the right data
- Select the right visualizations
- Keep your central message in mind



Building your central message

Describe in a few sentences the take-home message

1. Problem
2. Insights
3. Impact

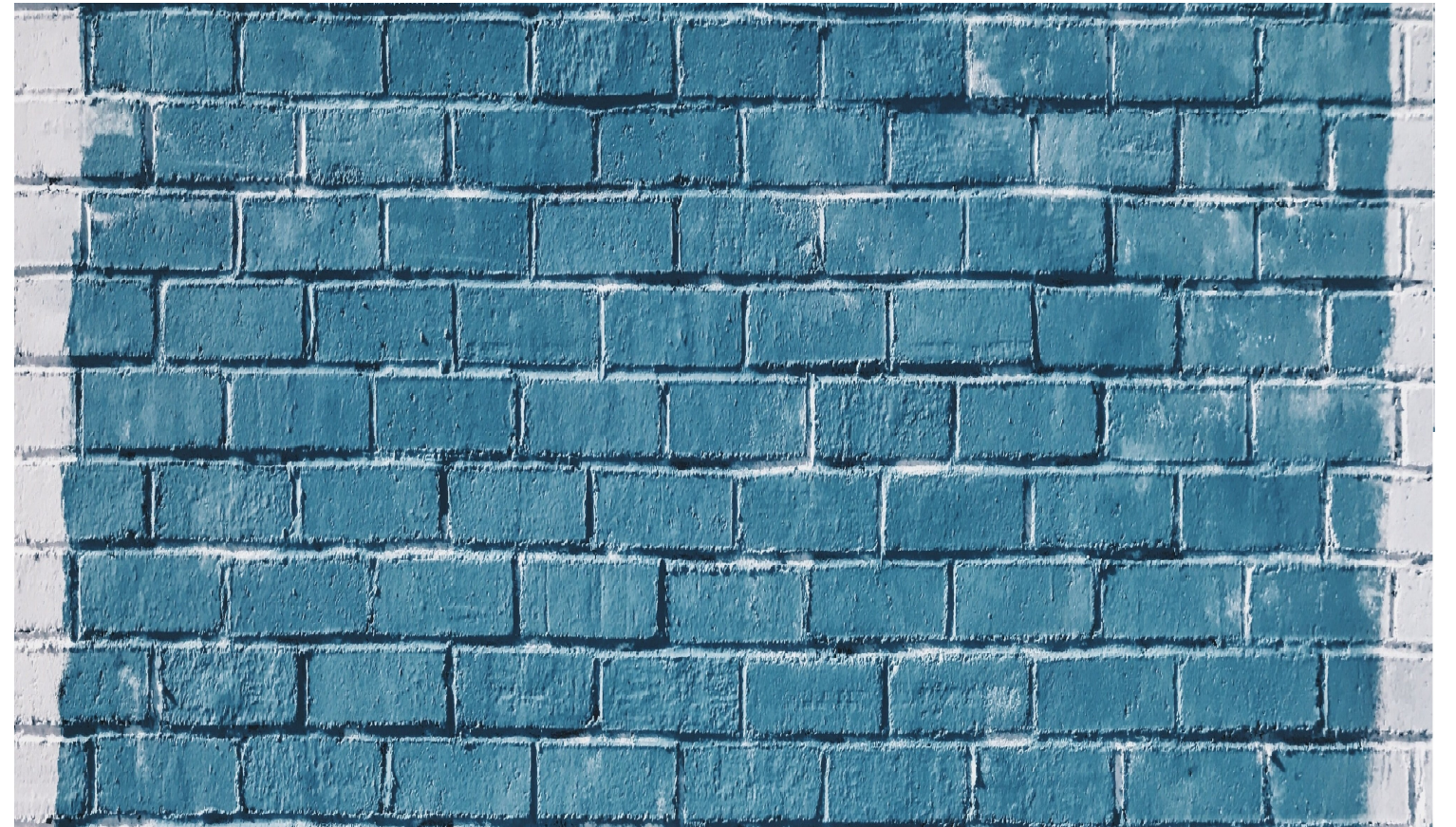
(Problem) Average math scores of students at University A are in decline for the last three years.

(Insights) Data analysis indicates that more students enroll with a weaker starting knowledge of mathematics.

(Impact) By organizing math summer schools, University A can help students better prepare for the expected knowledge level of mathematics.

Structure

- Use structuring techniques like storytelling and McCandless
- Think about visual structure in the lay-out
- Prepare an outline in advance and use this as a blueprint



Crafting an outline

1. Introduction
 - Data problem statement
 - Context
 - Objectives
2. Body
 - Data
 - Analysis
 - Key findings
3. Conclusions
 - Insights
 - Recommendations



Form

- Medium: oral vs. written communication
- Language: technical vs. non-technical
- Scope: short- vs. long-form

To find the appropriate format: think about the purpose of your communication and its audience



Thinking about your audience

- Who are they?
- What do they need to know?
- What do they want?

The University board:

- *Non-technical managers*
- *Need to know key findings*
- *Want specific recommendations*

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From data to story: a case study

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The case study: Customer Churn

- Problem: Customer churn has increased recently. Why?
- Analysis of customer churn rate over time and its characteristics
- Next task: presenting findings and recommendations



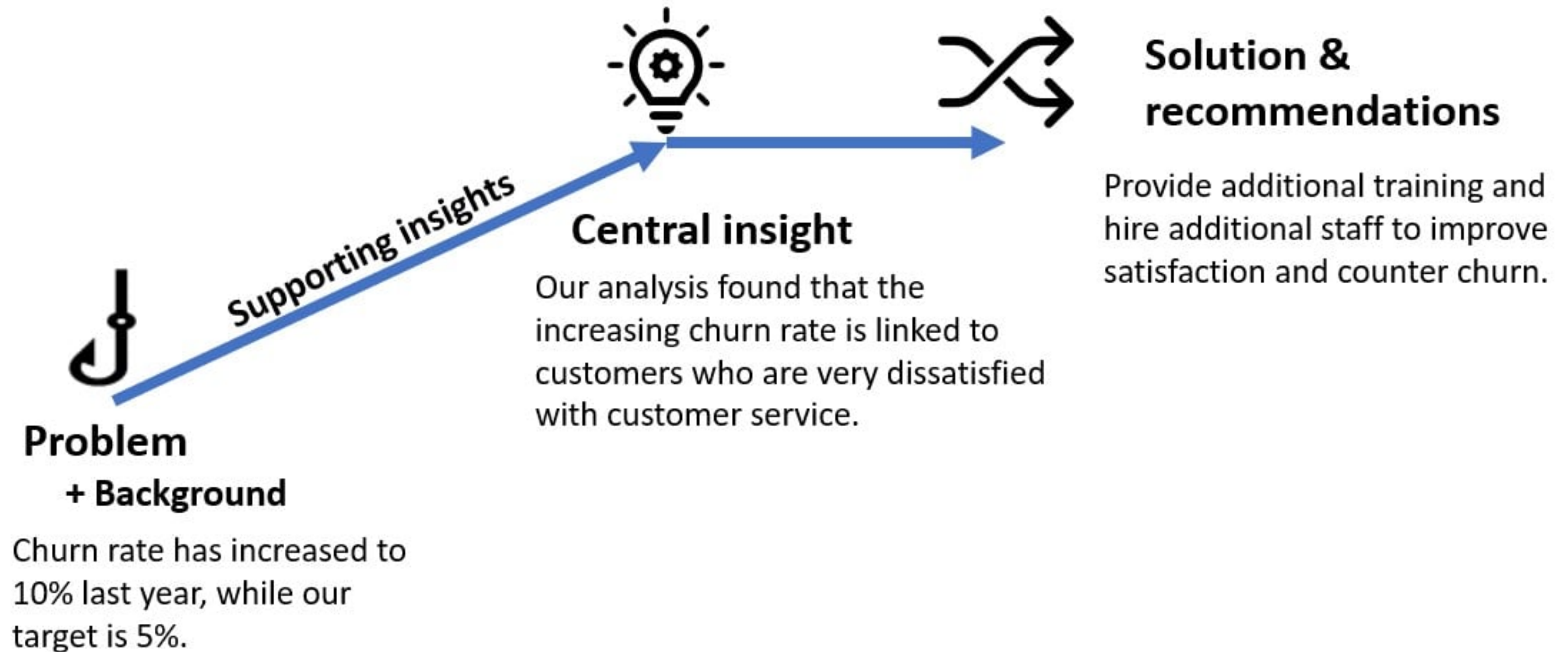
Formulating the central message

(Problem) The number of customers canceling their subscriptions has increased to 10% in the last year, while our target churn rate is 5%.

(Insights) Our analysis found that the increasing churn rate is linked to customers who are very dissatisfied with customer service.

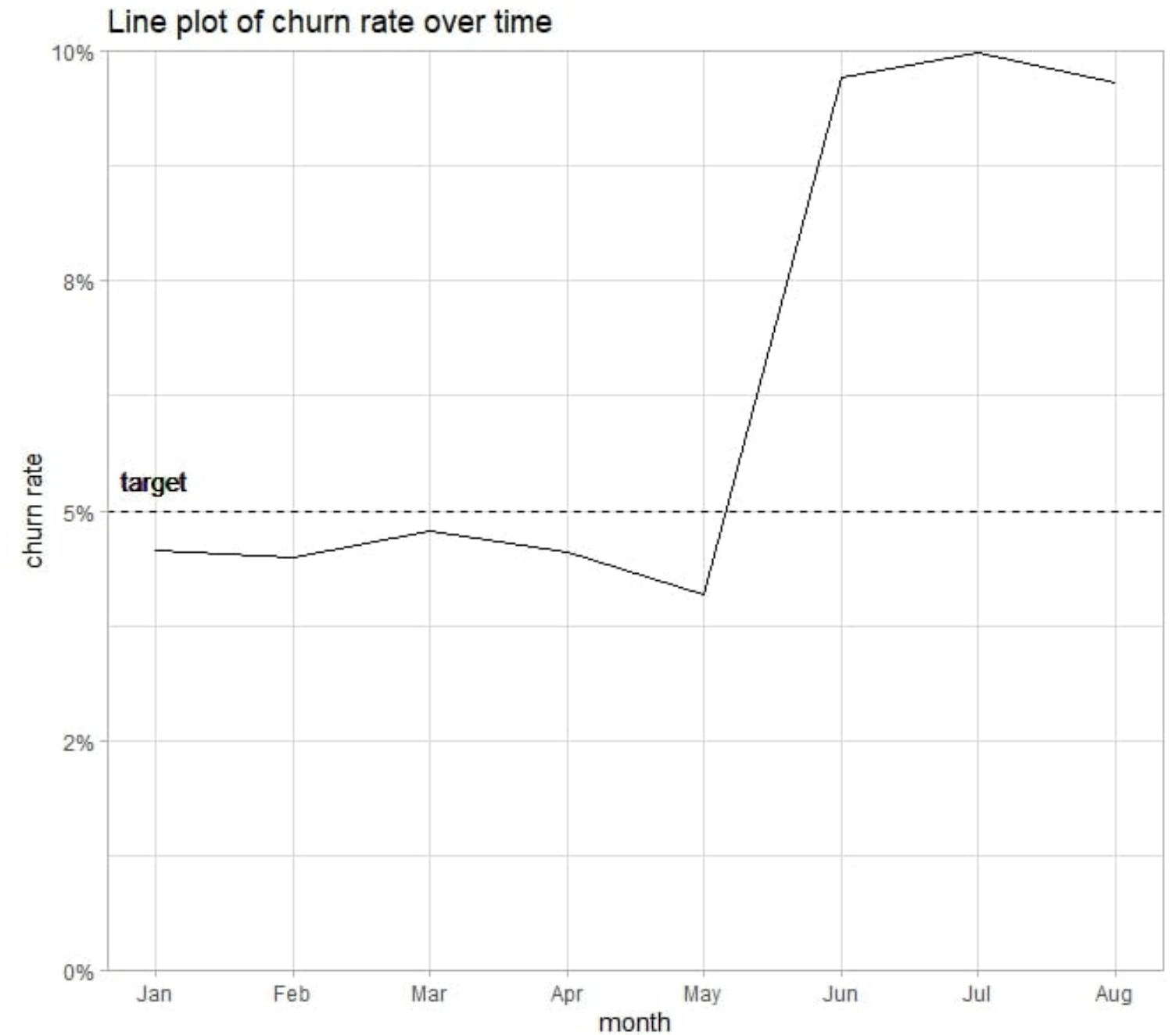
(Impact) By providing additional training for customer service staff and hiring additional staff to cope with the workload, we can increase customer satisfaction and counter the increasing churn rate.

Setting up the narrative structure



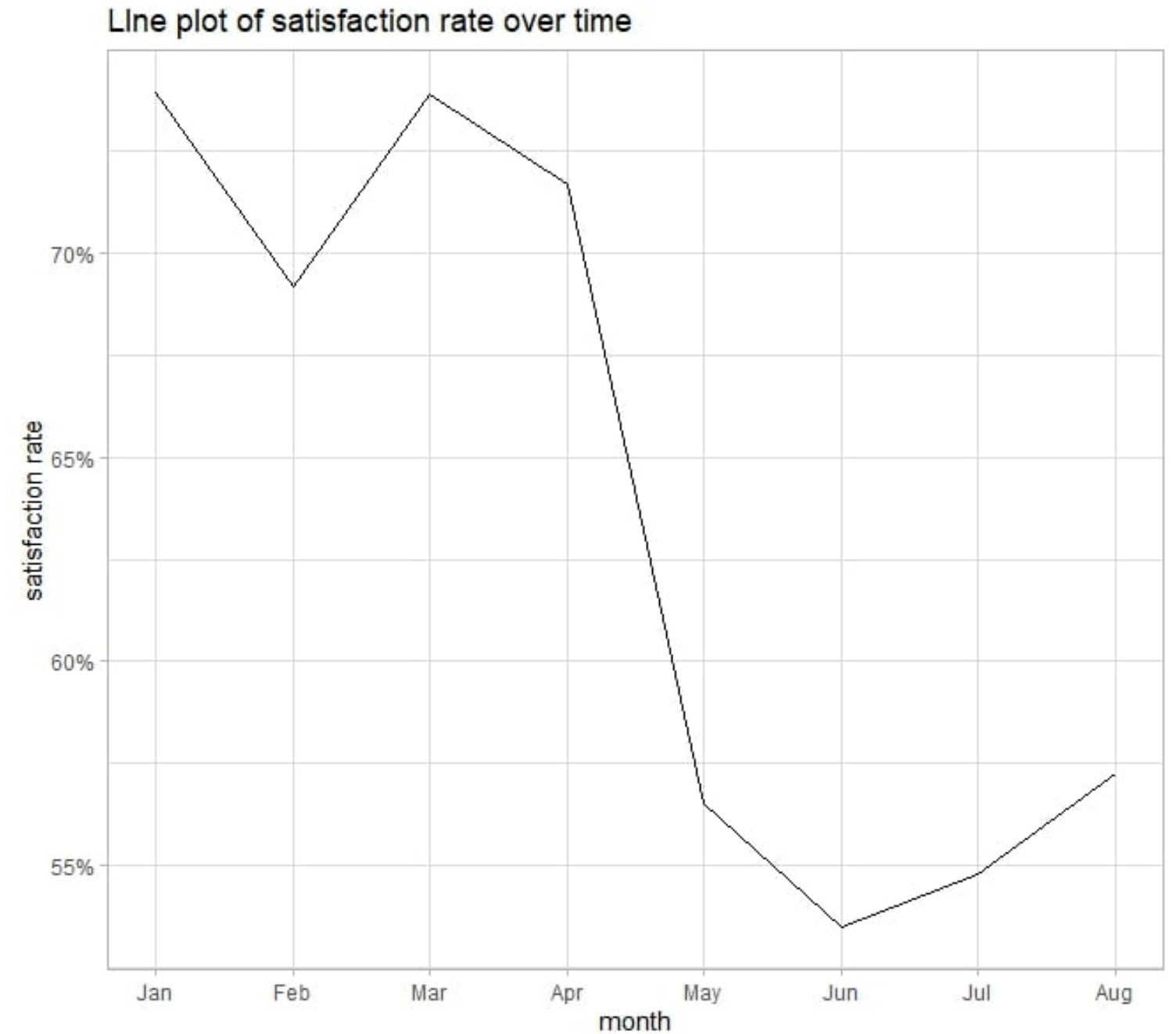
Choosing visualizations

- Problem: increasing churn rate



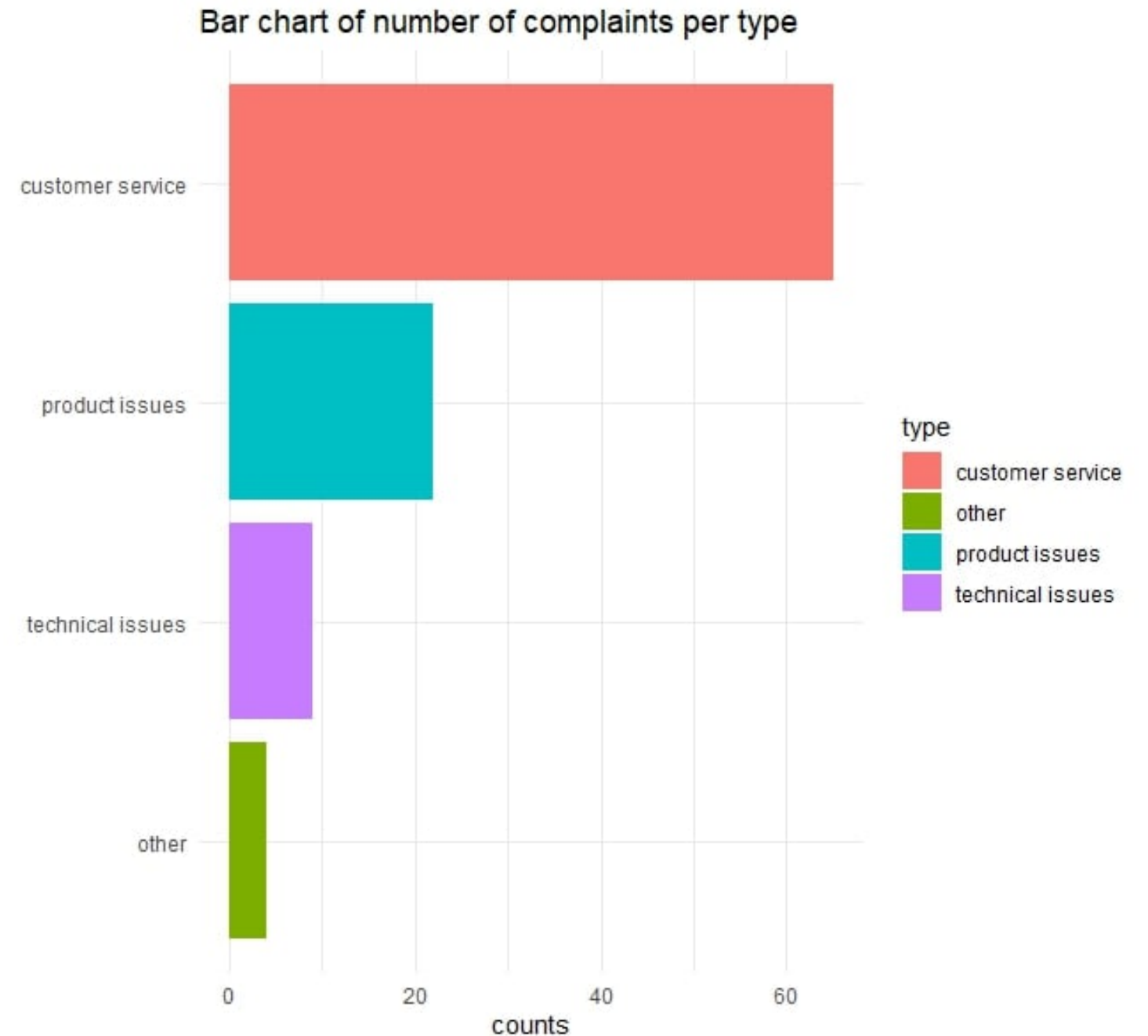
Choosing visualizations

- Problem: increasing churn rate
- Insight 1: increasing customer dissatisfaction



Choosing visualizations

- Problem: increasing churn rate
- Insight 1: increasing customer dissatisfaction
- Insight 2: majority of complaints are about customer service



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Wrap-up

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Recap

Chapter 1: Data literacy basics

- What it means to be data literate
- Why data literacy is an important life skill
- How to get from data to insights
- Data-driven decision making

Chapter 3: Getting insights from data

- The four types of analytics
- How to choose the right type of analytics for your question or business problem

Chapter 2: Reading data

- Where to find data
- How to distinguish different data types
- How to manage data
- How to handle common data problems

Chapter 4: Communicating insights

- How to use visualizations and storytelling
- The three keys to communicating effectively

Next steps

Suggested theoretical follow-up courses:

- Introduction to Statistics
- Understanding Data Science
- Understanding Data Visualization
- Understanding Machine Learning
- Data Communication Concepts

Suggested introductory courses:

- Introduction to Python
- Introduction to R
- Introduction to SQL
- Introduction to Power BI
- Introduction to Tableau

Congratulations!

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